



Aerodynamic Characterization of Wing-Wing Interactions for Distributed Lift Application

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Proximity effects due to wing-wing interactions were experimentally quantified as a function of gap and stagger across a wide range of different relative angles of attack (décalage). All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) on a pair of Clark-Y AR 2 semi-span wings. The proximity effects and the zone of influence at different gap and stagger locations were systematically characterized through measurement of the changes in aerodynamic force coefficients of individual wings and the combined wing-wing system. The wing angle of attack combinations that maintain similar aerodynamic efficiency at different gap and stagger locations were determined to allow for optimal placement of wings in a distributed lift system. A wide range of beneficial and detrimental combinations of gap, stagger, and décalage were determined.

I. Nomenclature

St	=	Stagger (Lateral Spacing)
G	=	Gap (Longitudinal Spacing)
α	=	Geometric Angle of Attack
$\alpha_{L=0}$	=	Zero Lift Angle of Attack
c	=	Chord
b	=	Wingspan
AR	=	Aspect Ratio; $AR = \frac{b^2}{S}$
U_∞	=	Freestream Velocity
C_L	=	Lift Coefficient; $C_L = \frac{Lift}{(1/2)*\rho*V^2*S}$
$C_{L\alpha}$	=	Lift Curve Slope
C_D	=	Drag Coefficient; $C_D = \frac{Drag}{(1/2)*\rho*V^2*S}$
L/D	=	Lift to Drag Ratio

Subscripts

1	=	Wing 1
2	=	Wing 2
sys	=	System

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II. Introduction

There has been a recent surge in the need for unmanned aerial vehicles (UAVs), drones, and air taxis for a variety of commercial, entertainment, and military applications. Aircraft designs put forth by companies such as Volocopter [1], Honda eVTOL [2], Ascendance [3], etc. have multiple lift producing surfaces and rotors acting in proximity to each other. These configuration choices are primarily informed by the “compactness” requirement in the design. For this reason, configurational choices are being considered that would otherwise not receive attention. Multi-wing configurations or distributed lift systems become a compelling choice in conceptual design of future UAVs and Private Air Vehicles (PAVs) that complements the vertical takeoff and landing capabilities of the design.

For multi-wing configurations to be considered in the early conceptual design process, the reliability of traditional lower order aerodynamic methods in predicting these aerodynamic effects must be determined. However, the nature of a highly distributed lift configuration, with 10 or more lifting surfaces in close proximity, does not lend itself to rapid or efficient viscous numerical solution. Moreover, highly distributed lift configurations drive individual lifting surface Reynolds numbers into a range where viscous interactions could have a profound effect on aerodynamic performance. As such, the degree of dependence of wing-wing interactions on viscous interactions could be determined in a first iteration through a reductionist approach. Focusing specifically on the three-dimensional viscous interactions and the aerodynamic forces on the upstream and downstream wings allows for a direct determination of the importance and isolated contribution of these effects.

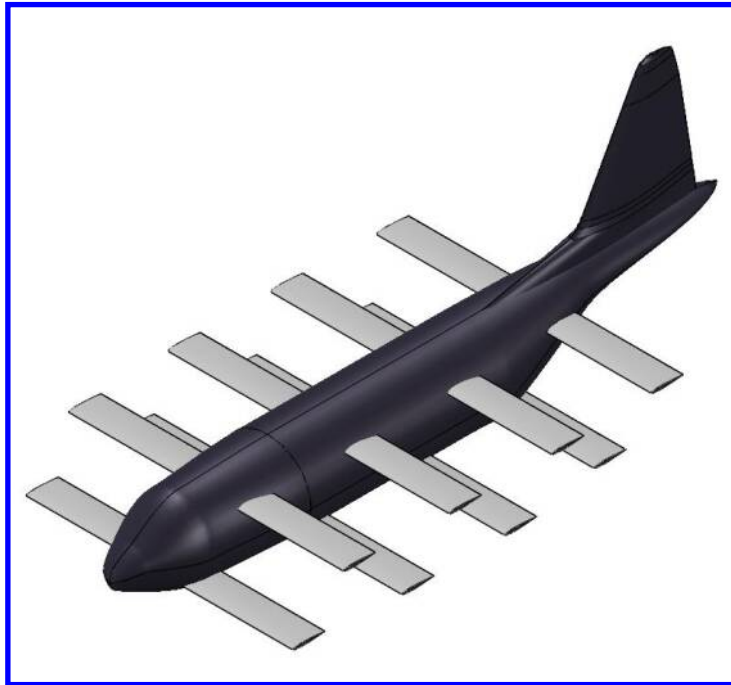


Fig. 1 OpenVSP Distributed Lift (Multi-Wing) Aircraft Concept

Quantifying interactions of lifting bodies dates back to Thwaites [4] who described the aerodynamics of two-dimensional biplanes and infinite arrays of lifting airfoils. Notable early contributions by the exact method of conformal transformation are attributed to Garrick [5]. Even though Söhngen inversion integrals provide an analytical way to determine the pressure distribution of a single-element airfoil in flow, for airfoils operating at numerous operating points with severe geometric and aerodynamic constraints and numerous degrees of freedom, the inversion integral method becomes increasingly difficult to apply. As such, quantifying interactions of lifting surfaces has predominantly been done using experimental and computational methods. Drela [6] developed an optimization algorithm for multi-point optimization of novel two-element transonic cruise airfoil with a thickness constraint and to maximize $c_{L_{max}}$ of a four-element landing configuration airfoil. Ragheb and Selig [7] performed computational investigations on seven multi-element airfoil configurations to replace the relatively thick airfoils used on modern megawatt-scale wind turbines. A plethora of other literature exists that quantifies the multi-element airfoil interactions in takeoff or landing configurations at a range of subsonic, transonic, and supersonic speeds. While these investigations

determine the optimal configurations of multi-element airfoils at extreme proximity to the main airfoil or wing, the chord fraction of the multi-element airfoil to the main airfoil is smaller. The present work focuses on multi-wing configurations where a collection of smaller wings is arrayed along the fuselage to generate lift that is conventionally generated by a mono-wing. A notional representation of a distributed lift system is shown in Fig. 1.

The distributed lift concept extends the biplane and the tandem wing configuration to include two or more wings in close proximity displaced in both lateral and longitudinal axes. Kang et al. [8,9] showed that in a biplane configuration, the gap and stagger had a large influence on performance with the upper wing producing more lift in all cases. But it was also capable of increasing the efficiency of the lower wing through effects of downwash. The study observed that beyond 1 chord gap and stagger the benefits of further wing-wing separation rapidly diminish.

Mongin et al. [10,11] studied various multi-wing configurations through both wind tunnel testing with a force balance and numerical investigations from Vortex Lattice Method codes. The wings were placed at fixed spacings and equal geometric angles of incidence therefore experiments were performed without adjusting inter wing angles (*décalage*). The results from this study showed that the multi-wing configurations were able to achieve around 50% of the aerodynamic efficiency, even approaching 61% in the case of the 6 Wing (3c Gap, 5c Stagger). Comparing the experiments to the VSPAERO potential flow code showed similar trends, but absolute values differed. This was hypothesized by the authors to be due to a lack of modeling of laminar separation bubble behavior and the method of wake resolution in the vortex method.

The emergence of multi-element designs along with prior studies concluding that there is potential in the distributed lift concept, leads to inter-wing optimization being a rational next step in exploring these configurations further. Understanding the wing-wing interactions allows evolved optimization of the configurations through strategic placement and relative angles of individual wings on the aircraft to achieve the desired customizable performance. Force and moment measurements are obtained to observe the performance of individual wings in a local reference frame which can then be related to the system in a global reference frame.

III. Experimental Setup

All experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) in the open-jet configuration. The tunnel has a 16:1 contraction ratio and a test section size of 76.2 cm by 76.2 cm by 137 cm with velocities ranging from 6.7 m/s to 36.7 m/s. The maximum turbulence intensity was measured to be 0.1% at 15 m/s by hot-wire.

The experimental testing campaign was conducted with two Clark Y AR 2 semispan wings at various gap and stagger locations as well as relative angles of attack (*décalage*). The center wing was fixed in the XY plane throughout the testing, henceforth called Wing 1. It was attached to a 6 component Analog ATI Gamma Net F/T sensor [12] which was mounted on top of a PDV PT-GD201 rotary stage [13] to adjust the angle of attack. Surrounding Wing 1 in the XY plane, a secondary wing, henceforth called Wing 2 was placed at 24 locations representing varying gap and stagger spacing relative to Wing 1. Wing 2 was attached to a 6 component Digital ATI Gamma Net F/T sensor which was mounted on top of a PDV PT-GD201 rotary stage to adjust the angle of attack and two FUYU FSK40 Linear Guides [14] to translate in the X and Y axis. A schematic of the experimental setup is shown in Fig. 2.

The ATI Gamma Net F/T sensor was orientated with the normal and axial forces measured through the X and Y axis. The sensing ranges of the Gamma in the SI-65-5 Calibration of X and Y is 65 N with a resolution of 1/80 N along with the torque range of 5 Nm and resolution of 1/1333 Nm. A sampling rate of 1000 Hz was used during data acquisition. The PDV GD-201 rotary stage instrument uncertainty in angle of attack was 0.01°. The FUYU FSK40 Linear Guide had an effective stroke of 600mm with a maximum horizontal and vertical load of 20kg and 10 kg. The precision of the linear guides was ± 0.03 mm.

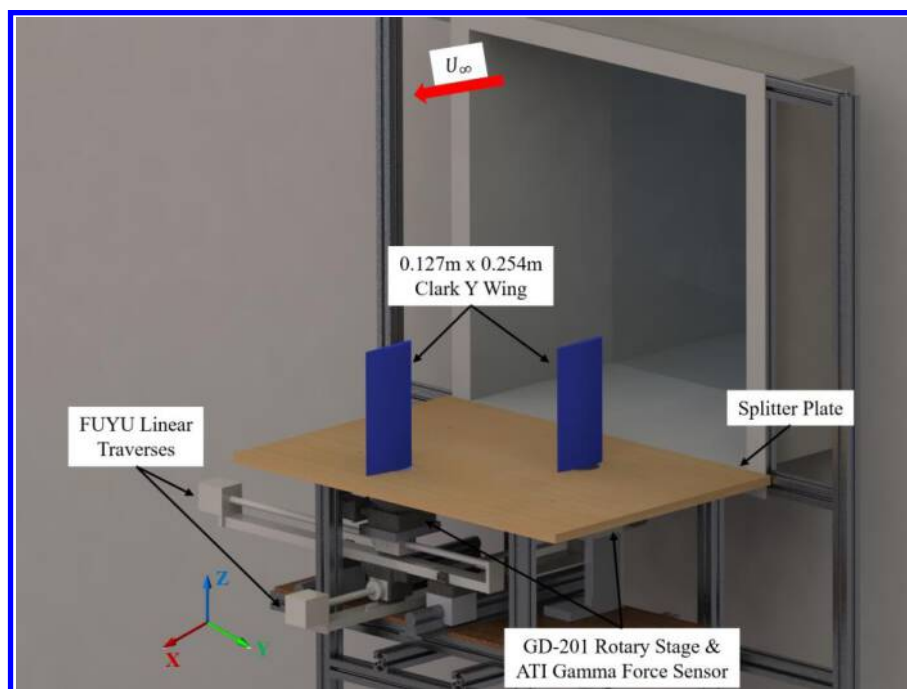


Fig. 2 CAD schematic of experimental setup in University of Dayton Low-Speed Wind Tunnel

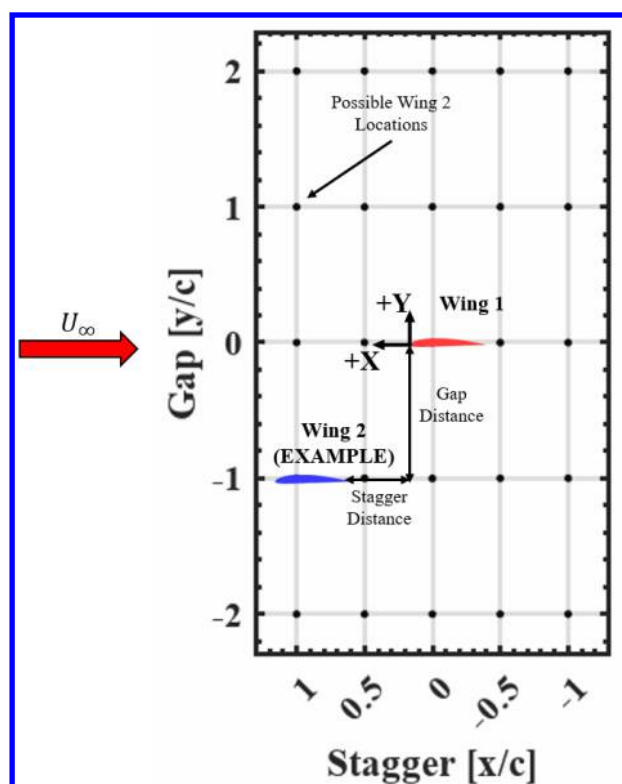


Fig. 3 Experimental nomenclature and Wing 2 grid locations

The coordinate system was centered on the leading edge of Wing 1. The naming convention for the locations is that the x-coordinate represents the stagger distance in chord between Wing 1 and Wing 2 and the y-coordinate is the

gap distance as annotated on the grid in Fig. 3 to remain consistent with biplane and previous Distributed Lift nomenclature.

Table 1 shows the test matrix of the experimental test campaign. The stagger and gap distances were investigated at 1 y/c and 0.5 x/c intervals respectively. Wing 2 was positioned at these gap and stagger locations at a fixed angle of attack while the Wing 1 angles were changed from -20° to 20°. Then the Wing 2 angle was changed at the same gap and stagger location and the Wing 1 angle of attack sweep is performed again. The same procedure was used at different gap and stagger locations shown in Fig. 3. The five Wing 2 angles of attack were selected to capture post negative stall, zero lift angle ($\alpha_{L=0}$), two intermediate attached flow angles, and then the point of maximum lift ($C_{L_{max}}$) based on a single Clark Y AR 2 wing. The baseline individual wing Reynolds number tested was 250,400 at a freestream speed of 30 m/s.

Table 1 Experimental Test Matrix

Reynolds Number	Stagger Distances [x/c]	Gap Distances [y/c]	Wing 1 Alpha Range [°]	Wing 2 Alpha Range [°]
250,400	1, 0.5, 0, -0.5, -1	2, 1, 0, -1, -2	-20 to 20	-10, -3, 8, 13, 20

IV. Results

The changes in performance of Wing 1 and Wing 2 in close proximity at different 2D spacing is discussed in this section. The 2D gap and stagger spacing with respect to the center wing are annotated in Fig 3 on the test grid above.

To compare performance of the wings in proximity with each other to a single wing baseline case, one of the Clark Y wings was tested separately at different positions in the spanwise direction and downstream of the wind tunnel inlet to ensure repeatability at various spatial locations in the test section. Figure 4 shows the averaged lift curve of the single wing from all the different tests along with the dashed lines representing the error bars. The baseline cases do not see a drastic stall in the angle of attack ranges tested but the lift plateaus at -10 degrees and starts to plateau at 15 degrees angle of attack. The lift curve slope (C_{L_α}) in the linear region of the averaged curve is calculated to be 0.057 deg⁻¹ which is around $C_{L_\alpha} = \pi\alpha$. This is consistent with low AR wing behavior. Using Helmbold's low aspect ratio equation [15] in Eq. (1), the aspect ratio corrected lift curve slope is calculated with C_{L_α} as 2π for thin airfoils and an effective AR of 3.1. The corrected lift curve slope was derived to be 0.0598 deg⁻¹ and is also plotted in Fig. 4 for comparison.

$$C_{L_\alpha} = C_{l_\alpha} \frac{AR}{\left(\frac{C_{l_\alpha}}{\pi}\right) + \sqrt{\left(\frac{C_{l_\alpha}}{\pi}\right)^2 + AR^2}} \quad (1)$$

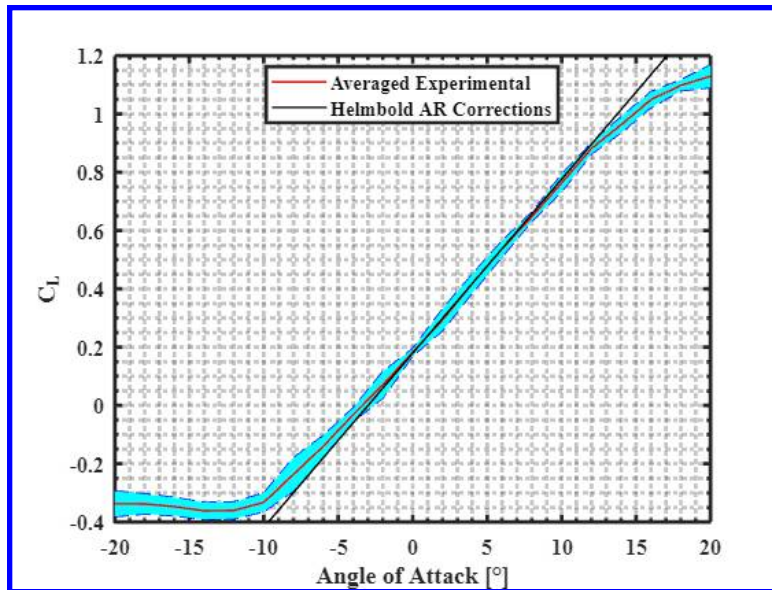


Fig. 4 Baseline AR 2 Clark Y Lift Curve

The system lift to drag ratio (L/D) was selected as the metric to quantify the overall wing-wing interactions. The system L/D for a specific configuration was calculated using the measured forces from the two wings added together as seen in Eq. (2). Figures 5-9 below show the three highest and lowest performing configurations compared to a baseline mono-wing represented as a solid black line. An Ideal System Efficiency as defined by Eq. (3) is provided to compare the extent of aerodynamic interactions when compared to the linear superposition of the lift and the drag forces. The efficiencies higher than that of the linearly superimposed ratio of the forces represents positive aerodynamic interference and vice-versa. Each subfigure represents Wing 2 locked at the stated angle of attack (α_2) as Wing 1 is pitched through an angle of attack sweep. The contours represent the system L/D or aerodynamic efficiency at a fixed Wing 1 and Wing 2 angle as annotated in the figures. The gap and stagger locations of the high and low performing configurations are also highlighted in the respective figures.

$$L/D_{sys} = \frac{L_1 + L_2}{D_1 + D_2} \quad (2)$$

$$Ideal\ System\ Efficiency = \frac{L_{Baseline\ Sweep} + L_{Baseline\ at\ \alpha_2}}{D_{Baseline\ Sweep} + D_{Baseline\ at\ \alpha_2}} \quad (3)$$

Across all of the Wing 2 sets, the best performing combined wing-wing cases can achieve between 30% and 130% of the isolated mono-wing max L/D . The worst performing cases range from 10% to 70% of the mono-wing efficiency. The large ranges of max L/D for both the high and low performers emphasize high sensitivity to the Wing 2 angle of attack. When Wing 2 is placed at -10° in Fig. 5, there is minimal difference between the highest performing and lowest performing cases. The minimum L/D occurs around -4° and maximum around 16° compared to the baseline mono-wing at -6° and 4° . All three high performers are below the mono-wing max L/D but do surpass the ideal system efficiency curve. Figure 6 has Wing 2 at -3° , the zero-lift angle of attack for a Clark Y ($\alpha_{L=0}$), where the difference between the high and low performing configurations become prominent. Increments in system L/D are visible at high Wing 1 angles of attack for the high performing configurations. The three highest system L/D cases happen at a Wing 2 location at negative gap and low stagger where they all perform very similarly. The lowest aerodynamic efficiency results from the tandem configurations.

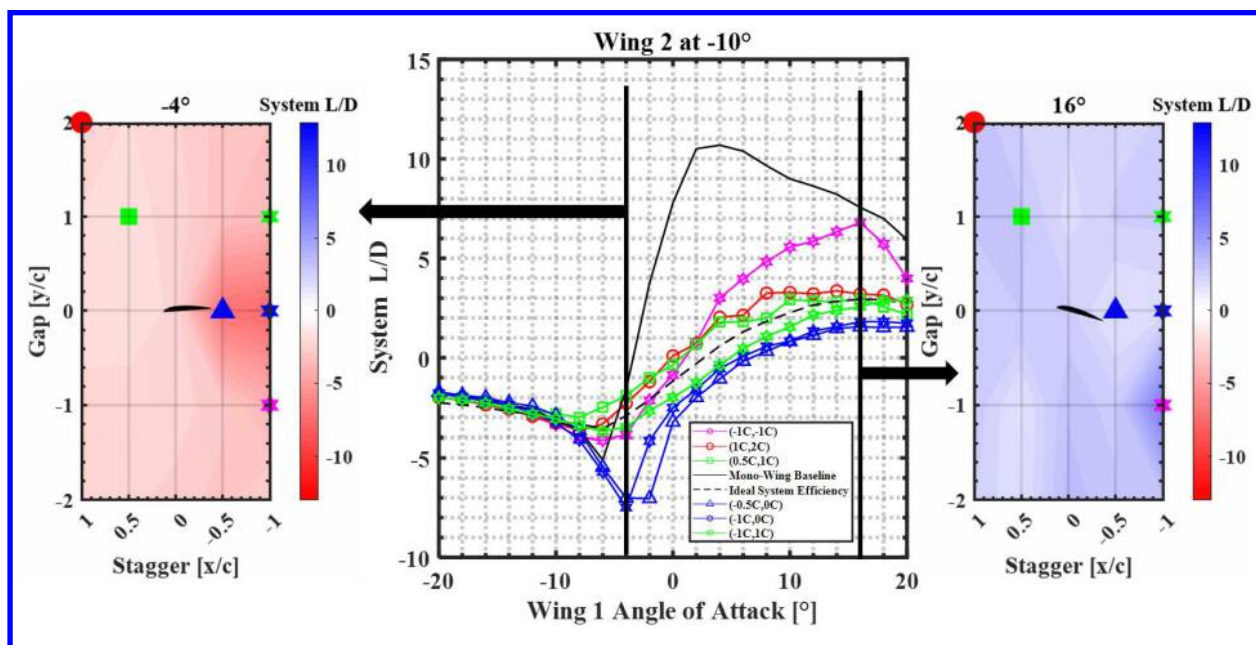


Fig. 5 System L/D of Various Configurations as Wing 1 sweeps and Wing 2 is fixed at -10°

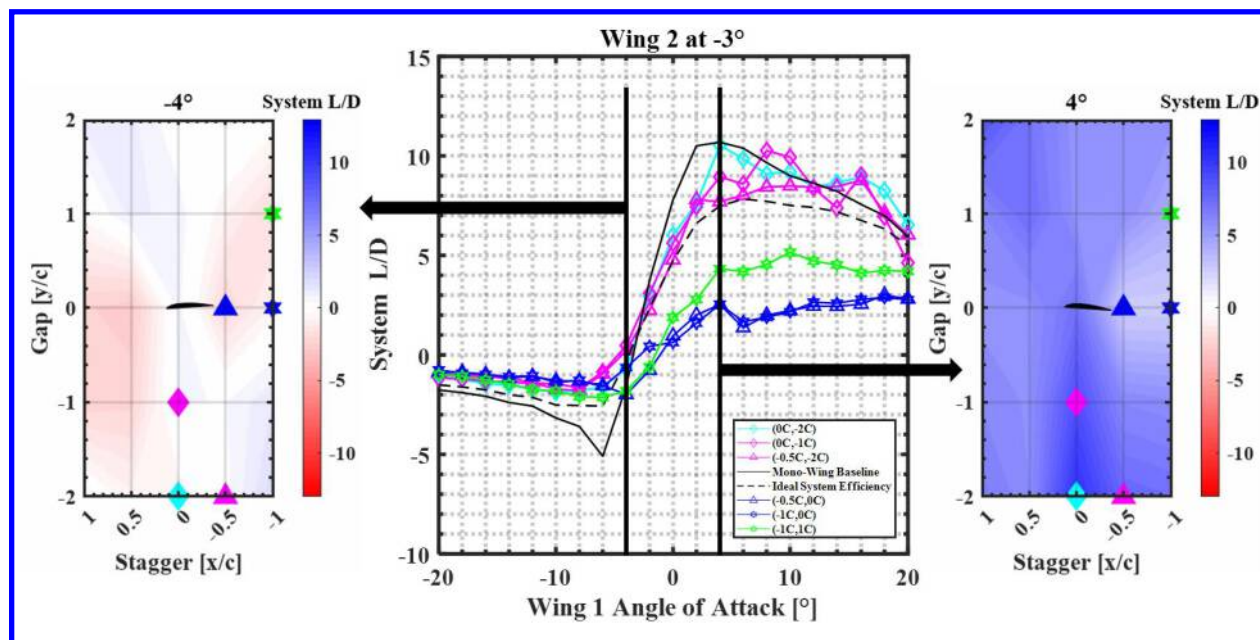


Fig. 6 System L/D of Various Configurations as Wing 1 sweeps and Wing 2 is fixed at -3°

As α_2 increases to 8° in Fig. 7 and 13° in Fig. 8, the two-wing configuration's efficiency approaches that of the mono-wing and also becomes positive at negative Wing 1 angles of attack. This is an artifact of the relatively fixed contribution to lift of Wing 2 (at a fixed angle) at high angles of attack overcoming lift decrements from Wing 1 at negative angles of attack. The isolated baseline Clark Y lift curve produces a minimum C_L of around -0.3 at -20° and surpasses this magnitude of C_L value by 4° . This results in a fully positive system C_L when Wing 2 is above 4° . The ideal system efficiency makes a similar transition at the negative Wing 1 angles of attack. The combined wing-wing configurations surpass the maximum efficiency of the mono-wing and ideal two-wing system when Wing 2 is at a high positive angle of 8° and 13° . A notable trend is a gradual shift of the maximum lift to drag ratio to lower Wing 1 angles of attack with increment in Wing 2 angle of attack.

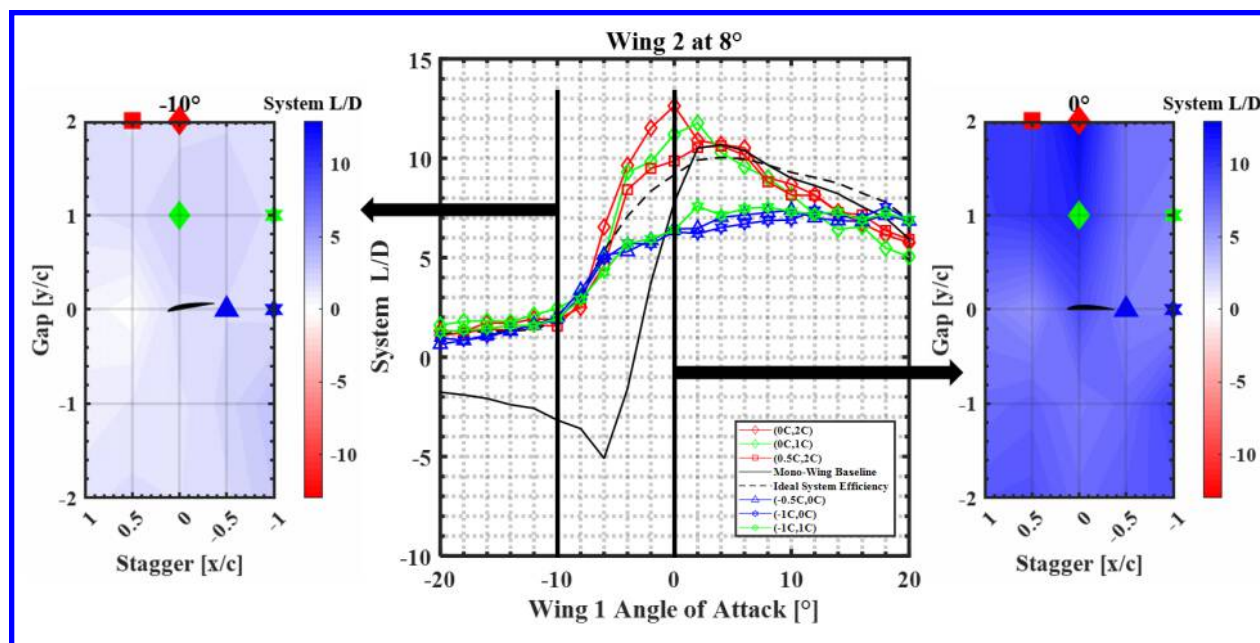


Fig. 7 System L/D of Various Configurations as Wing 1 sweeps and Wing 2 is fixed at 8°

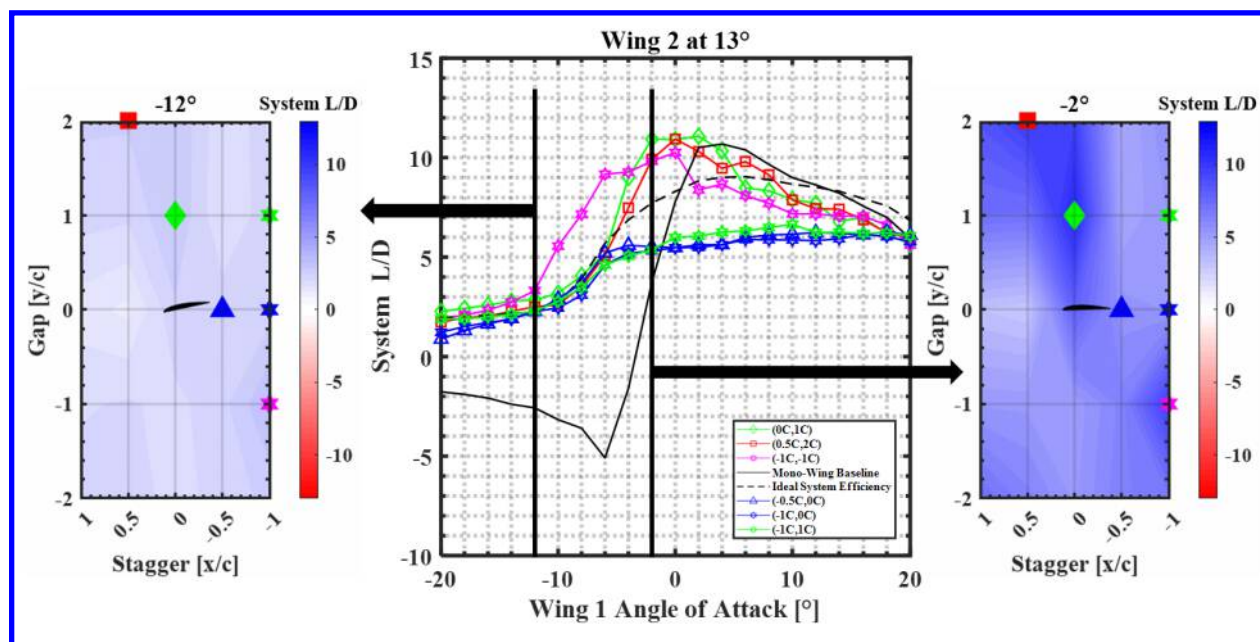


Fig. 8 System L/D of Various Configurations as Wing 1 sweeps and Wing 2 is fixed at 13°

Increasing α_2 to 20° in Fig. 9 causes a decrement of max L/D possibly due to Wing 2 approaching stall. The L/D curve begins to plateau with no drastic max L/D angle of attack. Wing 2 at -20° introduces one new high performer along with two new low performance configurations resulting in the biggest change when the high and low performers were fairly constant at the previous Wing 2 angles of attack. Overall, low stagger and positive gap locations of Wing 2 seem to result in optimal performance. Three configurations are constant throughout $\alpha_2 = -10^\circ, -3^\circ, 8^\circ$, and 13° as being the poorest performing configurations. Two of those three are tandem wing configurations where the wings are horizontally coplanar.

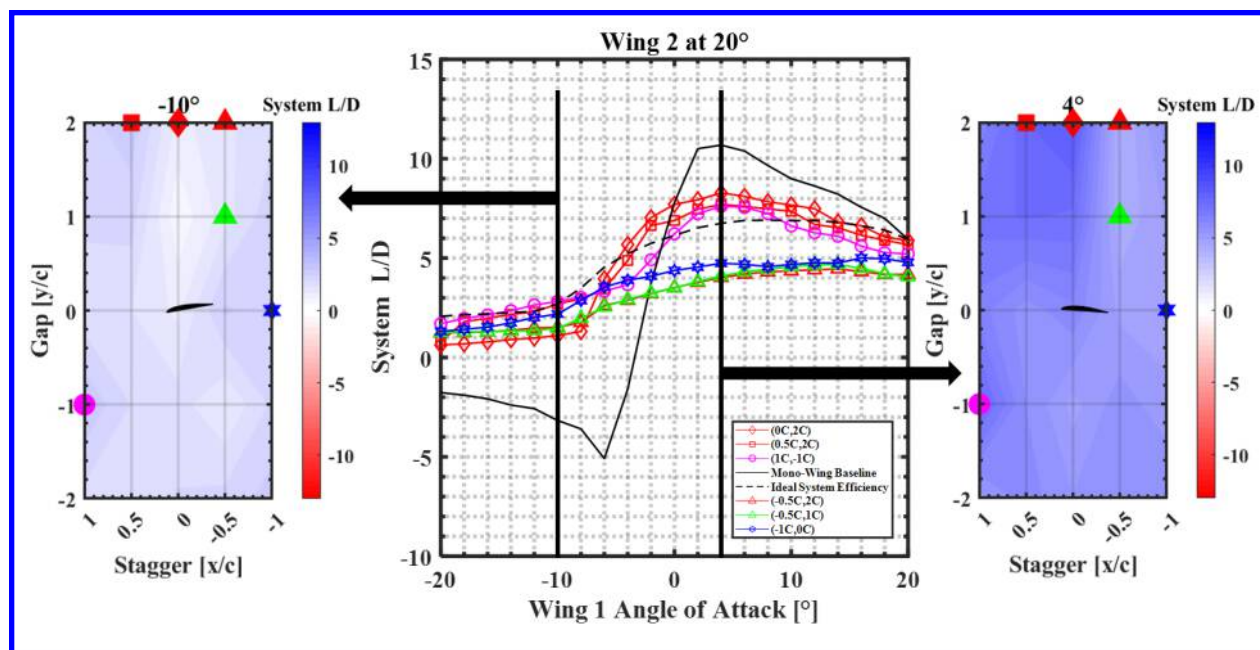


Fig. 9 System L/D of Various Configurations as Wing 1 sweeps and Wing 2 is fixed at 20°

The individual wing contributions can be compared to understand how a specific wing relative position can lead to an efficiency comparable to that of the mono-wing. Wing 2 placed at a 0.5c Stagger and 2c Gap consistently performs higher than the mono-wing at several different α_2 's. Figure 10 shows the lift coefficient and drag coefficient of both wings as Wing 1 goes through an angle of attack sweep and Wing 2 is locked at 8°. The standalone baseline coefficient results were plotted for comparison as well. Wing 1 experiences a lower C_L and a higher C_D when compared to the baseline. The C_L of Wing 2 however remains around the same value of C_L as experienced by the baseline at 8°. It does see a slight increment when Wing 1 is at a negative angle of attack and then a decrement as Wing 1 moves to a positive angle. The Wing 2 C_D however experiences a significantly lower C_D when compared to the baseline wing. This decrement in Wing 2 C_D appears to be the cause of the two-wing system L/D surpassing that of the mono-wing. This could possibly be reasoned through a mirror vortex type effect via downwash deflection.

The lowest performing case can also be studied using a similar approach to understand the individual wing contributions to the system L/D. Irrespective of the α_2 , the classical pure tandem configurations, represented by a 0c Gap, are categorically the lowest performing configurations. Below in Fig. 11 the performance of Wing 2 placed 1c downstream of Wing 1 at $\alpha_2 = 8^\circ$ is offered as a representative example. Wing 1 and Wing 2 perform lower than the baseline case in C_L at most angles of attack, but the C_D of both wings show a mixed behavior. At negative angles of attack, Wing 1 has an increased drag and Wing 2 has a lower drag which can be reasoned through a wake blockage-type effect. As the angle of attack increases however, Wing 1 experiences lower relative drag while Wing 2 peaks and then drops below the baseline drag. This large variation in drag and large drop in lift for Wing 2 is hypothesized to be from direct interference effects and induced downwash from Wing 1 (as it is swept in angle of attack) directly upstream of Wing 2. This is reflected in the drastic decrement in Wing 2 C_L at higher Wing 1 angles of attack.

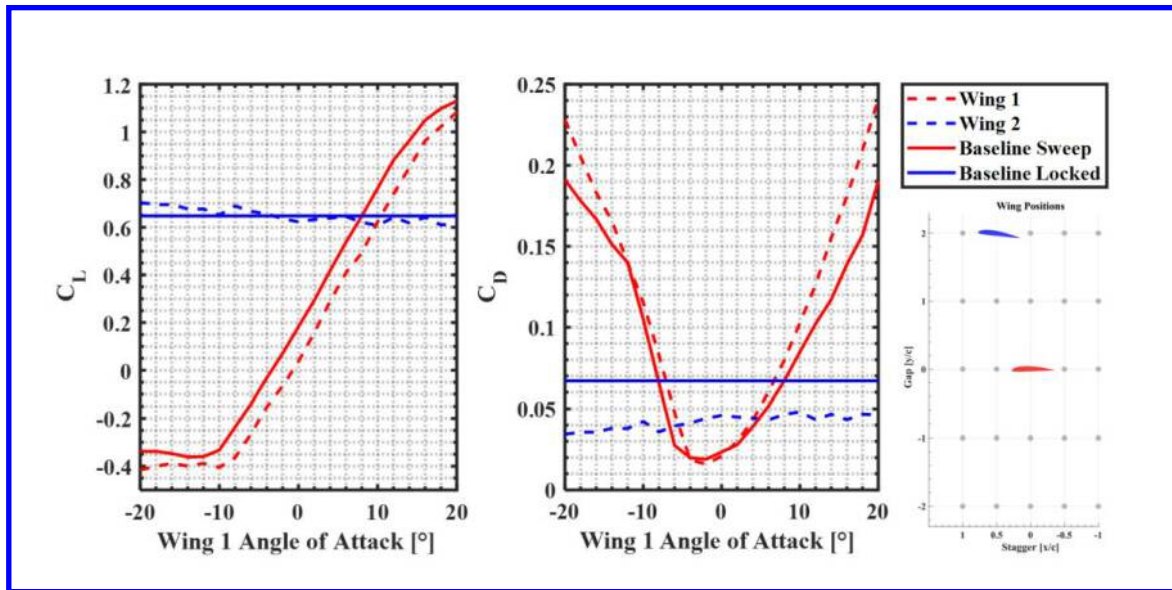


Fig. 10 Lift and Drag Coefficient of Wing 1 and Wing 2 at Position (0.5c St, 2c G) locked at 8° compared to the Baseline

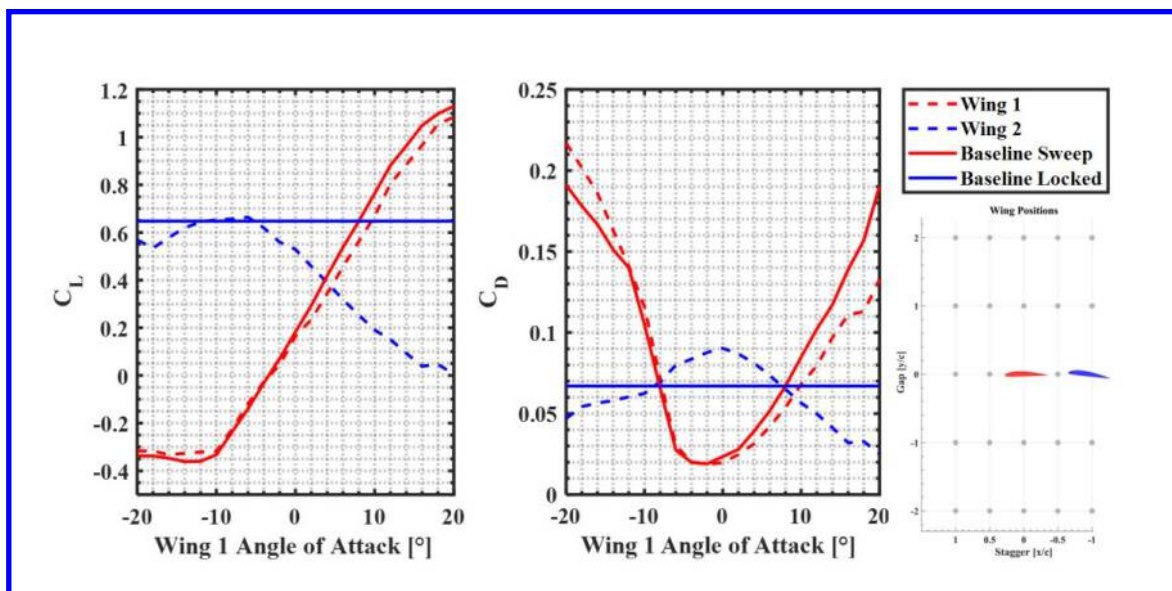


Fig. 11 Lift and Drag Coefficient of Wing 1 and Wing 2 at Position (-1c St, 0c G) locked at 8° compared to the Baseline

With a view towards applied optimization of the air vehicle performance the system L/D is once again leveraged. In vehicle design, L/D has a strong effect on range and endurance performance. The Bréguet Range and Endurance Equations in Eq. (4) and Eq. (5) use the L/D along with other aircraft parameters to predict the total range (distance) and the total endurance (time) [16]. Knowing the impact on L/D of a distributed lift configuration can be useful in the design of an aircraft.

$$Range = \int_{w_i}^{w_f} \frac{U_\infty}{c} \frac{C_L}{C_D} \frac{1}{W} dW \quad (4)$$

$$Endurance = \int_{W_i}^{W_f} \frac{1}{c} \frac{C_L}{C_D} \frac{1}{W} dW \quad (5)$$

The isosurface plots in Fig. 12 represent a specific target L/D selected based on the aircraft designer's requirements. The standalone baseline mono-wing has an L/D of 10.7 at 4° as shown in Fig. 5 through 9. Attempting to achieve approximately 80% of this efficiency with an L/D of 8, the possible angle of attack configurations of a two-wing system are displayed through the plots below. Increasing the target L/D above 80% significantly diminishes the possible configurations, while decreasing below 80% allows most of the configurations to achieve the target. Therefore 80% is set as the highest target with a large variety of possible designs.

The isosurface plots indicate that 80% of the mono-wing efficiency can be obtained at a selected stagger and gap locations but is heavily influenced by the angle of attack of combinations of Wing 1 and Wing 2. When Wing 2 is at -3°, a L/D of 8 could only be achieved in negative gap locations at a Wing 1 angle of attack range from 0° to 10°. When Wing 2 is at 8° however, the same efficiency can be achieved across almost the entire stagger and gap landscape, except for the tandem wing configurations.

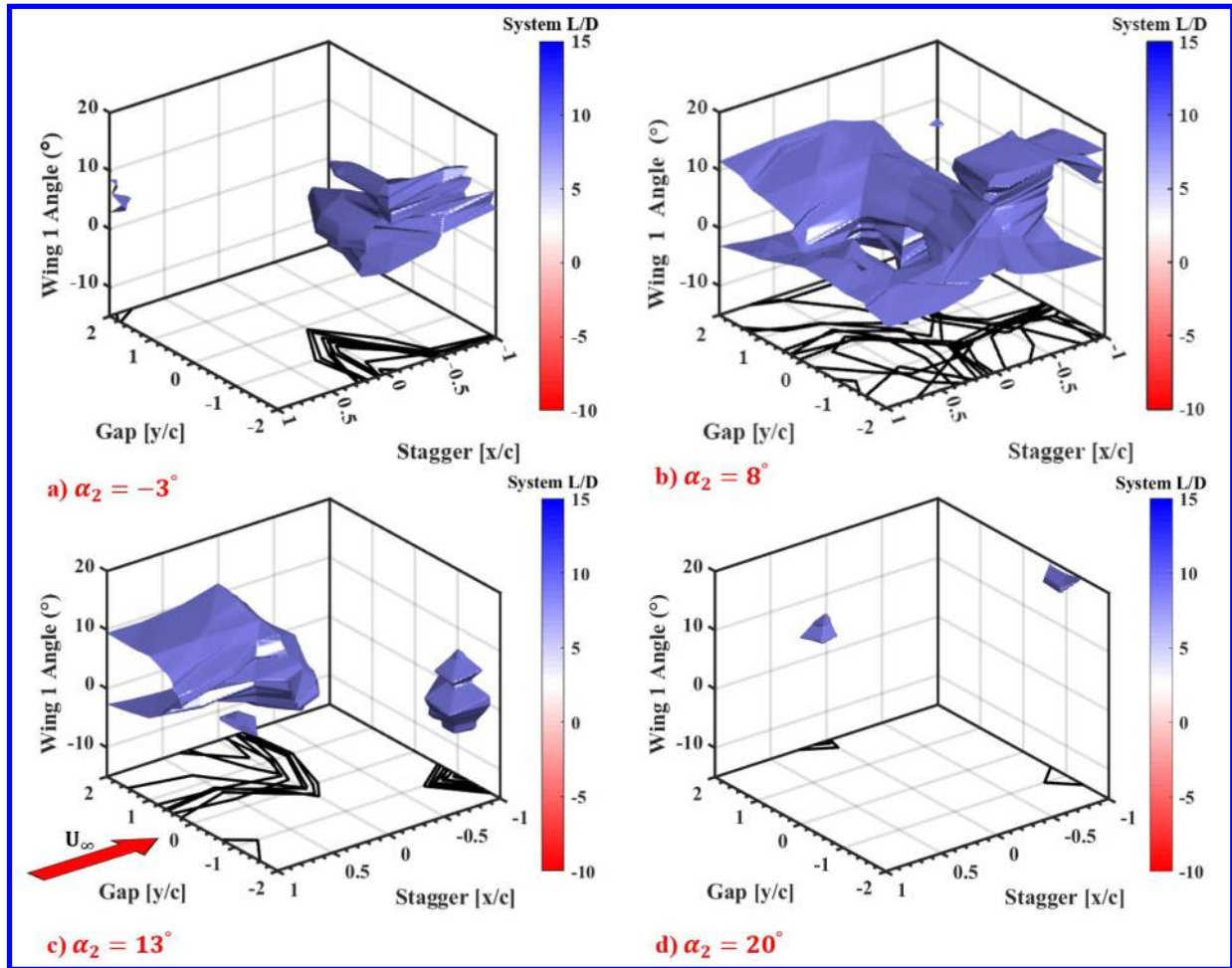


Fig. 12 Stagger, Gap, and α_1 combinations capable of achieving Target L/D with Wing 2 at a) -3°, b) 8°, c) 13°, d) 20°

The gap and stagger locations at which the same efficiency can be obtained reduces at higher Wing 2 angles of attack as shown in Fig. 12c and 12d. At these higher Wing 2 angles of attack, Wing 1 is also at a higher angle of attack to achieve the same efficiency and is thus susceptible to a higher degree of interference due to stall onset. Most of the tandem wing configurations cannot achieve this target as seen in the L/D plots shown above. There is a significant drop in efficiency for the tandems compared to the mono-wing and other two-wing configurations.

With individual wing forces and moments being measured from the experiments, the influence of the wings on each other can be explored in a more focused approach. The effect of Wing 1 on Wing 2 and vice-versa can be represented in a 2D space (gap and stagger) at selected Wing 1 and Wing 2 angles of attack. Figure 13 is a visual representation of the different stagger and gap positions of Wing 2 overlaid onto the ΔC_L Contour with $\alpha_1 = -20^\circ$ and $\alpha_2 = 20^\circ$ below as a representative easily visualized example for what is displayed in the figure arrays below in Fig. 14 and 15.

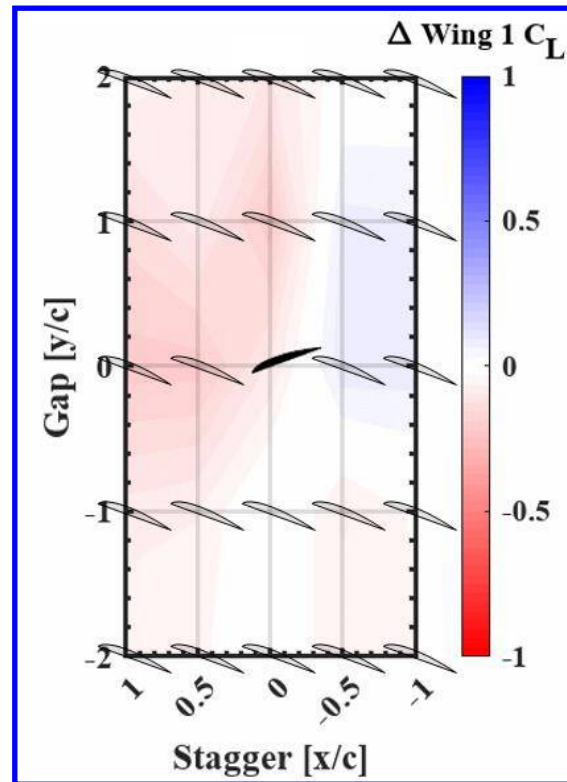


Fig. 13 Visual Representation of Wing Locations on Contours with Example of ΔC_L Contour with $\alpha_1 = -20^\circ$ and $\alpha_2 = 20^\circ$

The contours in Fig. 14 represent the Wing 1 ΔC_L determined by subtracting the baseline C_L from the Wing 1 C_L at different gap and stagger locations. Throughout all the different Wing 1 and Wing 2 angle of attack combinations, Wing 1 mostly experiences a decrement when Wing 2 is placed at any of the 24 positions around it. This decrement becomes increasingly prominent at greater Wing 1 and Wing 2 angles in the bottom right of the figure array. For the tandem configuration, specifically with Wing 2 upstream, Wing 1's lift coefficient drops significantly when compared to the baseline. When Wing 2 is at -10° , the opposite is true as Wing 1 experiences a slight increment in C_L . Minimal effect in C_L is observed when Wing 2 is placed downstream as it does not measurably affect the flow experienced by Wing 1. Wing 2 at its $\alpha_{L=0}$ (-3°) in general causes only small Wing 1 C_L variance from the baseline at all positions compared to other α_2 values.

Contours of Wing 2's ΔC_L in Fig. 15 show slightly different trends as compared to Fig. 14. Just as Wing 1 saw decrements in C_L when Wing 2 was placed upstream, Wing 2 sees decrements when Wing 1 is placed upstream with some inconsistencies. Increments are witnessed at low α_1 and α_2 where $\alpha_1 = -20^\circ$ and -10° with $\alpha_2 = -10^\circ$, -3° , and 8° . This occurred for Wing 1 but less frequently as it was only observed when $\alpha_1 = -10^\circ$, -2° , and 8° with $\alpha_2 = -10^\circ$. Around zero lift angle of attack (-2° or -3°), the influence of both wings on each other is at its minimum. Both Fig. 14 and 15 do show that even with both wings close to $\alpha_{L=0}$, there are slight increments and decrements at select locations due to physical localized disruptions in the flow resulting from the presence of the companion wing.

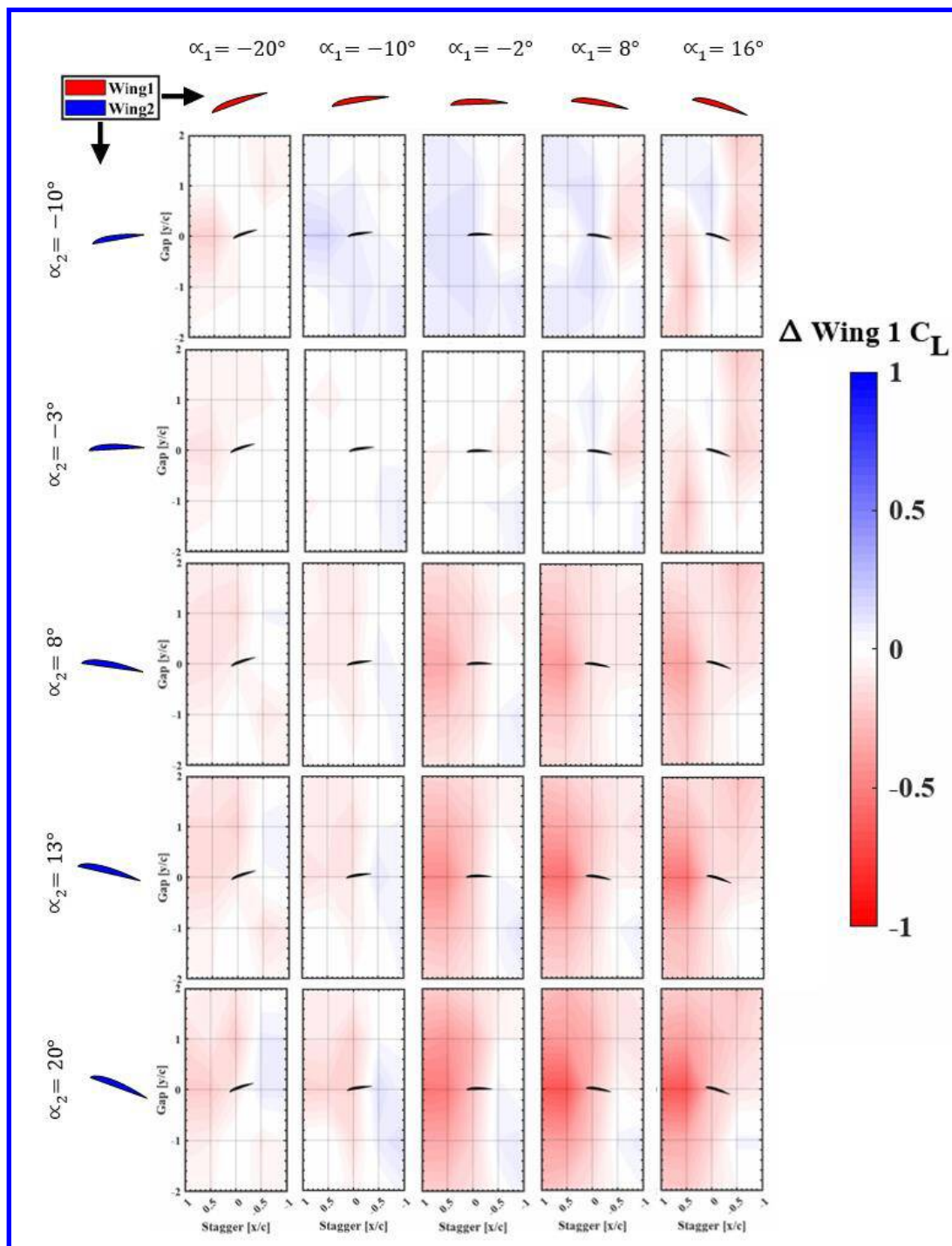


Fig. 14 Contours of Difference in Wing 1 C_L compared to Baseline C_L at different Wing Alpha Combinations

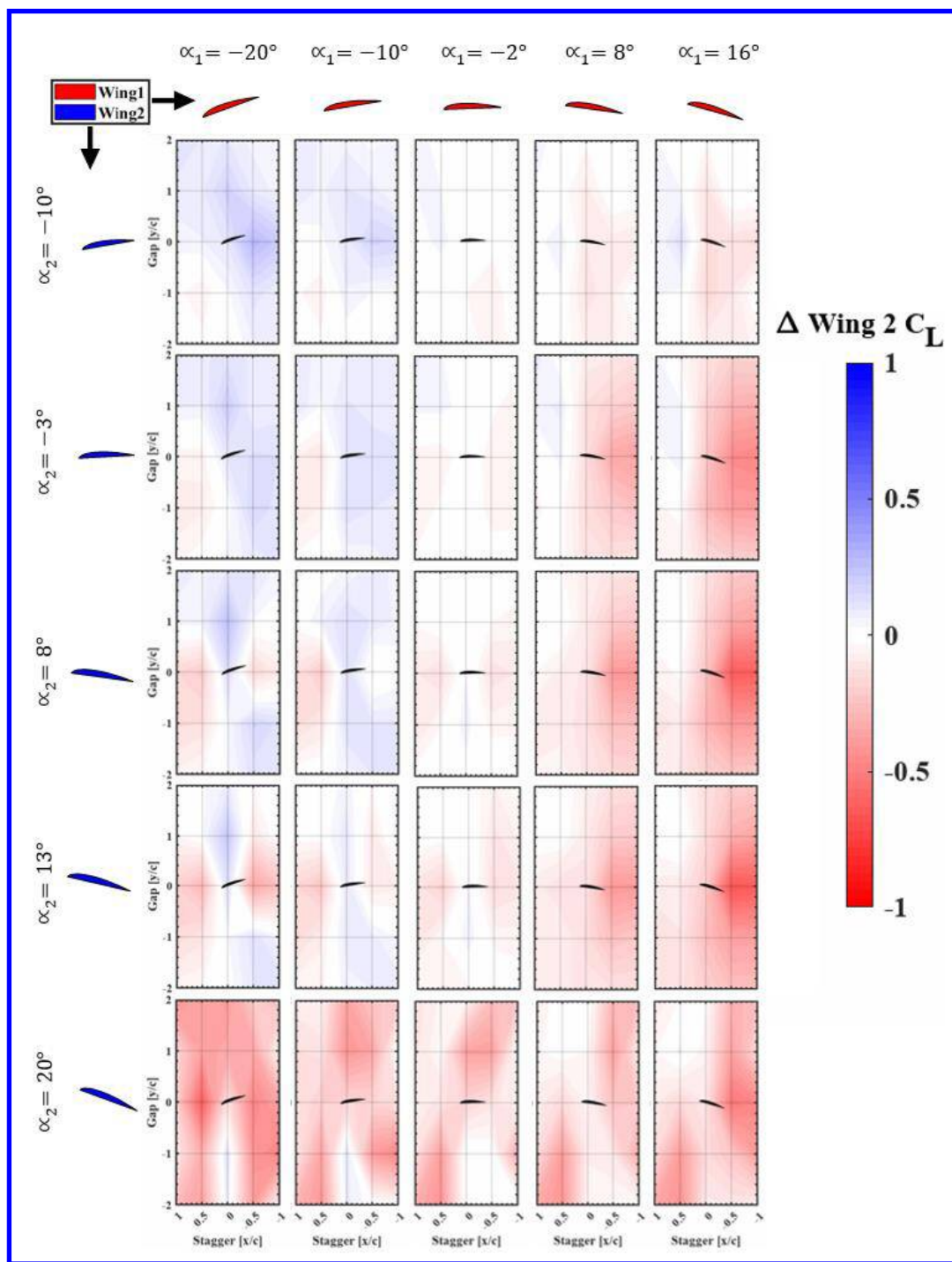


Fig. 15 Contours of Difference in Wing 2 C_L compared to Baseline C_L at different Wing Alpha Combinations

Accounting for all the interaction effects and the influences of the wings on each other at all angles of attack, the overall mean system L/D was obtained at each stagger and gap distance from Wing 1 as shown in Fig. 16. As evidenced through the aforementioned figures, the tandem configurations have the lowest mean L/D obtaining values below the baseline mean L/D of around 3.5. When compared to the mean ideal system efficiency that was calculated in Eq. 2, there are no two-wing configurations that see a system L/D increment. The lowest decrements in system L/D occur at the top left and bottom right corner where the combined stagger and gap combination is large and interactions are at

a relative minimal. Also noticeable is the slight asymmetry which increases the importance of the direction of stagger and gap of Wing 2 respective to Wing 1.

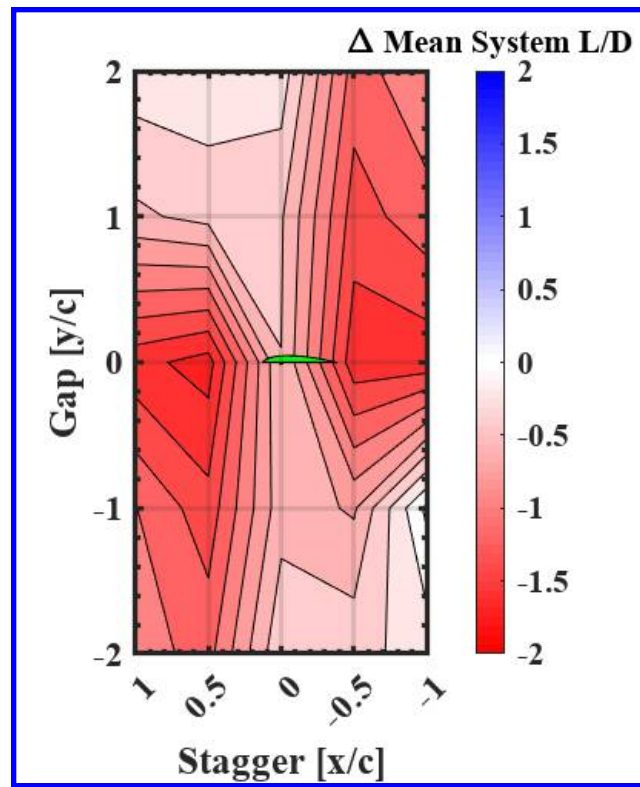


Fig. 16 Difference in Mean System L/D from Mean Ideal System Efficiency across all angles of attack at various gap and stagger locations representing Zone of Influence

V. Conclusions

The degree to which placing a wing in proximity to another impacts aerodynamic efficiency was parametrically explored using experimental investigations at different gap, stagger, and *décalage* angles. The results indicated that a system L/D higher than that of the mono-wing baseline could be achieved with a two-wing configuration. This can be done at several different relative wing positions and angle of attack combinations. Additionally, shifts in maximum L/D angle of attack were observed as a function of certain angle of attack, gap, and stagger configurations and could be used to tailor overall vehicle performance in flight as a function of changes in load.

Separating the systems into the individual wing contributions enables the influence of the wings on each other to be quantified. The areas of influence were also determined through contours of individual wing performance when another wing is placed in close proximity. Understanding the proximity effects will allow more informed optimization for distributed lift configurations with variable numbers of wings. Most notably, the decision to place the distributed wings in tandem at zero *décalage* angle will be revisited. Leveraging the results presented here could have a profound impact on improving the aerodynamic performance of the previously tested distributed lift configurations.

Hence one promising area of future work is to increase the number of wings tested in the present array and determine whether the benefits remain when a full system of wings is implemented. Further studies with particle image velocimetry can provide information about the underlying flow physics responsible for the differences in system-level performance documented here.

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